Math 261-001
Quiz 9 Solutions, November 6

1. **Final answers:**

$$\boxed{\text{(a) Line integral of } \vec{F}}, \qquad \boxed{\text{(b) Surface integral of } \operatorname{curl}\vec{G}}.$$

(a) Stokes' theorem replaces the flux of the curl through the hemisphere by the line integral of the original field along its boundary:

$$\iint_S (\operatorname{curl}\vec{F})\cdot\vec{N}\,d\sigma = \oint_{\partial S}\vec{F}\cdot\vec{T}\,ds.$$

The boundary is the circle $x^2+y^2=4$, $z=0$. The outward orientation of the upper hemisphere induces counterclockwise traversal as viewed from above. Thus the replacement is a *line integral of* $\vec{F}$, not a line integral of its curl.

(b) Stokes' theorem replaces the line integral by the surface integral of the curl:

$$\oint_C \vec{G}\cdot\vec{T}\,ds = \iint_{S_T}(\operatorname{curl}\vec{G})\cdot\vec{N}\,d\sigma.$$

One suitable surface S_T is the flat triangular sheet in $x+y+z=2$ with the three stated vertices. The traversal $(2,0,0)\to(0,2,0)\to(0,0,2)\to(2,0,0)$ induces an upward normal. Thus the replacement is a *surface integral of* $\operatorname{curl}\vec{G}$, not the flux of $\vec{G}$ itself.

Neither part requires differentiation or evaluation. The type of integral changes when passing to the other side of Stokes' theorem.

2. **Final answers:**

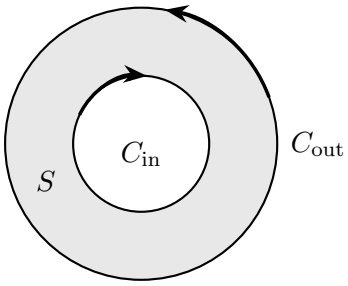
$$\boxed{\begin{array}{c}\text{Outer: } x^2+y^2=4,\ z=0;\\ \text{counterclockwise from above}\end{array}} \qquad \boxed{\begin{array}{c}\text{Inner: } x^2+y^2=1,\ z=0;\\ \text{clockwise from above}\end{array}}.$$

The annulus has two boundary components. With the unit normal $\vec{N}=(0,0,1)$ pointing upward, the annular surface must stay on the left as each component is traversed.

This gives counterclockwise traversal along the outer circle and clockwise traversal along the inner circle. The edge of the hole is part of the boundary just as the outer circle is.

Equivalently, positively oriented parametrizations are

$$\begin{aligned}\vec{c}_{\text{out}}(t) &= (2\cos t, 2\sin t, 0),\\ \vec{c}_{\text{in}}(t) &= (\cos t, -\sin t, 0),\end{aligned} \qquad 0\le t\le 2\pi.$$



View from above; $\vec{N}=(0,0,1)$.

3. **Final answer:** $\boxed{3}$.

Write the plane as $z = 2 - 2x - y$ and parametrize the sheet by

$$\vec{r}(x, y) = (x, y, 2 - 2x - y), \qquad \begin{cases} 0 \le x \le 1, \\ 0 \le y \le 2 - 2x. \end{cases}$$

Its tangent vectors give

$$\vec{r}_x = (1, 0, -2), \qquad \vec{r}_y = (0, 1, -1), \qquad \vec{n} = \vec{r}_x \times \vec{r}_y = (2, 1, 1).$$

The positive third component gives the required upward orientation. The unit normal is $\vec{N} = (2, 1, 1)/\sqrt{6}$ and $d\sigma = \sqrt{6}\, dx\, dy$, so $\vec{N}\, d\sigma = (2, 1, 1)\, dx\, dy$.

On the surface,

$$\vec{F}(\vec{r}(x, y)) = (y, x, y + 2 - 2x - y) = (y, x, 2 - 2x),$$

and hence

$$\vec{F}(\vec{r}) \cdot (\vec{r}_x \times \vec{r}_y) = 2y + x + 2 - 2x = 2 + 2y - x.$$

Therefore

$$\begin{aligned} \iint_S \vec{F} \cdot \vec{N}\, d\sigma &= \int_0^1 \int_0^{2-2x} (2 + 2y - x)\, dy\, dx \\ &= \int_0^1 \left[(2 - x)y + y^2\right]_0^{2-2x} dx \\ &= \int_0^1 (8 - 14x + 6x^2)\, dx \\ &= \left[8x - 7x^2 + 2x^3\right]_0^1 = 3. \end{aligned}$$

4. **Final answer:** $\boxed{6}$.

For $\vec{F} = (z - x^{2026}, x - y^{2026}, y - z^{2026})$, the curl is

$$\operatorname{curl} \vec{F} = (1 - 0,\ 1 - 0,\ 1 - 0) = (1, 1, 1).$$

Choose the flat triangular sheet with vertices $P = (2, 0, 0)$, $Q = (0, 2, 0)$, and $R = (0, 0, 2)$. Since

$$(Q - P) \times (R - P) = (-2, 2, 0) \times (-2, 0, 2) = (4, 4, 4),$$

the stated traversal $P \to Q \to R \to P$ matches an upward normal. Parametrize the triangle by

$$\vec{r}(x, y) = (x, y, 2 - x - y), \qquad \begin{cases} 0 \le x \le 2, \\ 0 \le y \le 2 - x. \end{cases}$$

Then $\vec{r}_x \times \vec{r}_y = (1, 1, 1)$. By Stokes' theorem,

$$\begin{aligned} \oint_C \vec{F} \cdot \vec{T}\, ds &= \iint_{S_T} (\operatorname{curl} \vec{F}) \cdot \vec{N}\, d\sigma \\ &= \int_0^2 \int_0^{2-x} (1, 1, 1) \cdot (1, 1, 1)\, dy\, dx \\ &= \int_0^2 \int_0^{2-x} 3\, dy\, dx = \int_0^2 (6 - 3x)\, dx = 6. \end{aligned}$$

The large powers disappear because only derivatives with respect to the other components' variables enter the curl.